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Industrial Internship Report on

" Forecasting of Smart city traffic patterns"

Prepared by

Rakshitha K N

Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time. My project was "Forecasting of Smart City Traffic Patterns." The project focuses on analyzing historical traffic data from four city junctions to understand traffic patterns and forecast the number of vehicles using Machine Learning. The project includes data preprocessing, feature engineering, exploratory data analysis, holiday and working-day traffic analysis, peak-hour identification, and comparison of different Machine Learning models. Among the models evaluated, Gradient Boosting achieved the best performance with an R^2 score of 0.9691. An interactive Streamlit dashboard was also developed to visualize traffic patterns and provide traffic forecasts. This internship gave me a valuable opportunity to gain practical exposure to Data Science, Machine Learning, data analysis, visualization, and application development. It helped me understand how a real-world problem can be transformed into a data-driven solution and improved my practical knowledge through hands-on project development. Overall, it was a valuable learning experience that enhanced my technical skills and understanding of Machine Learning projects. .

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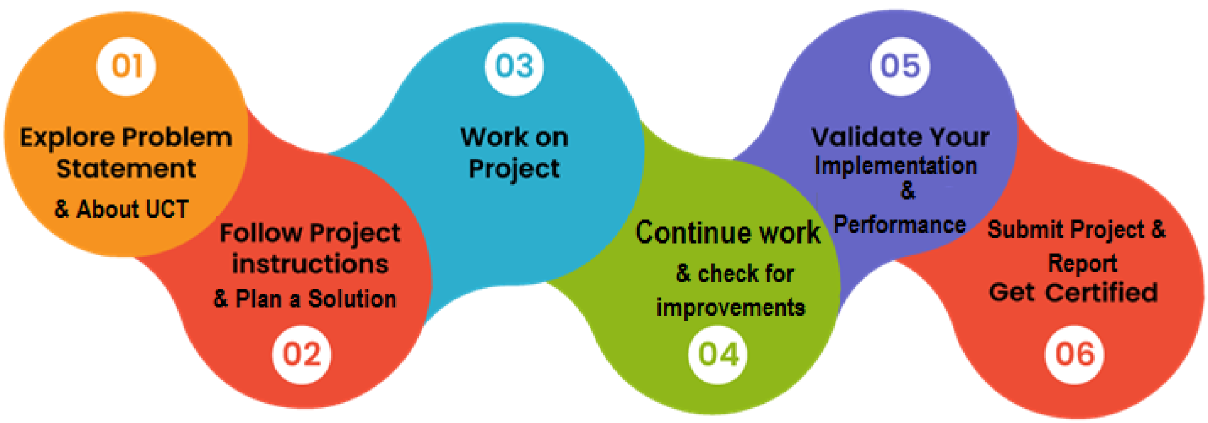
1 Preface

This six-week Industrial Internship provided me with practical exposure to Data Science and Machine Learning and helped me apply my academic knowledge to a real-world problem.

My project, "Forecasting of Smart City Traffic Patterns," focuses on analyzing traffic data from four city junctions, identifying traffic patterns, peak hours, holidays, and working-day variations, and forecasting traffic using Machine Learning.

The project was completed through stages including data preprocessing, feature engineering, EDA, model training, evaluation, and Streamlit dashboard development.

This internship helped me improve my knowledge of Python, Data Analysis, Machine Learning, data visualization, and application development. Overall, it was a valuable learning experience that helped me understand the complete process of developing a practical Data Science project.



2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

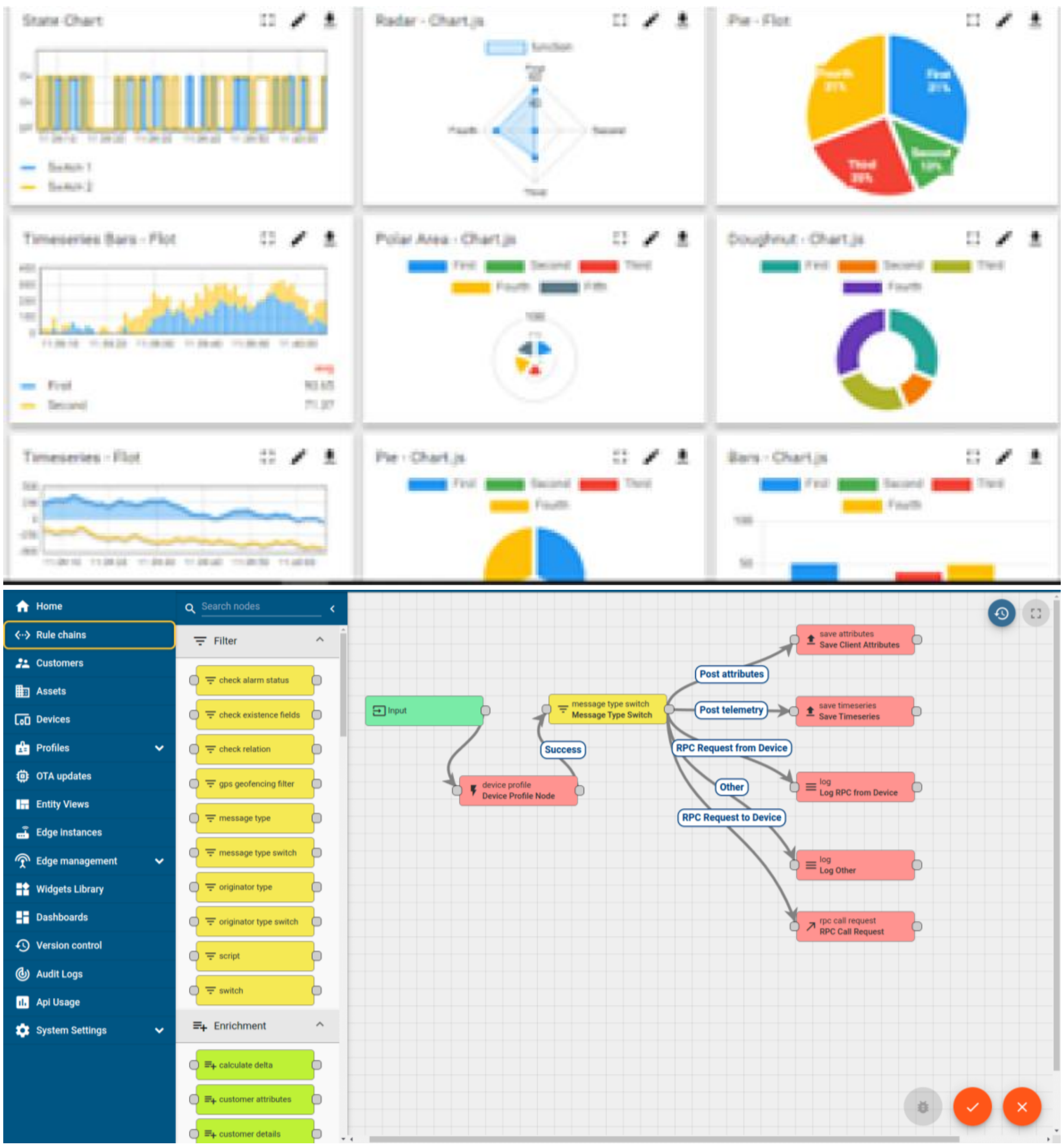
i. UCT IoT Platform (uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

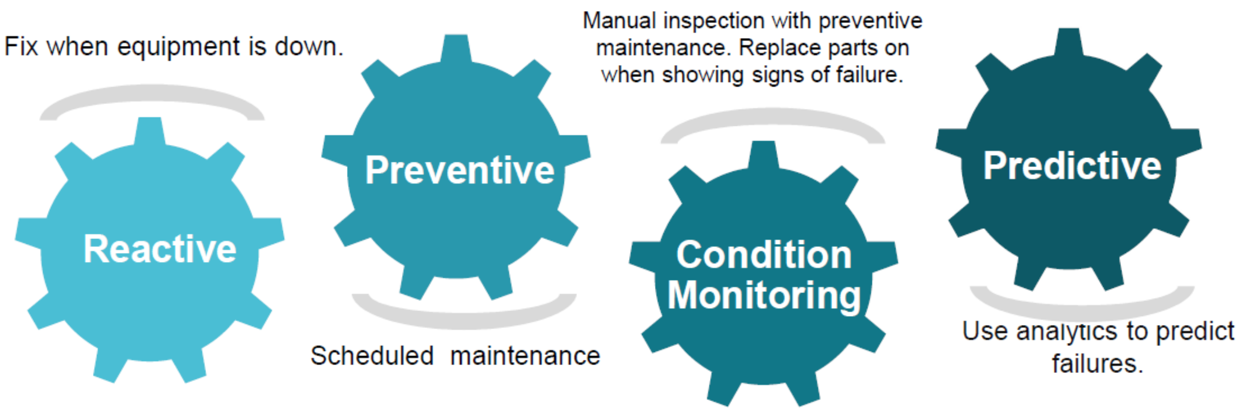
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

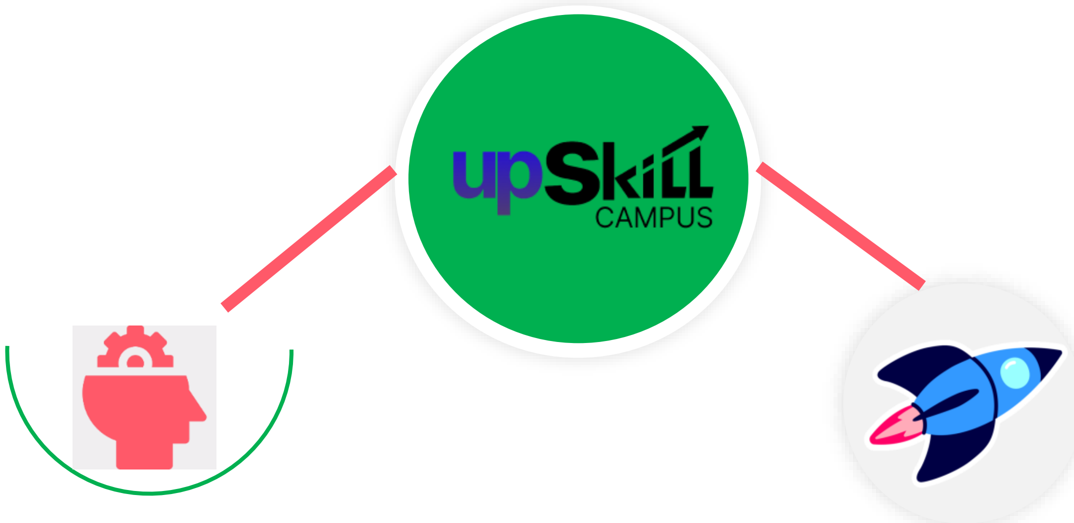
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/

Career growth/upskilling
- Interview Preparation and skill building
- upskilling Courses
- Skill Assessment
- Profile building

Professional networking
- Alumni Connections
- Mentorship
- Discussion/QA forum

Collaboration platform
- Project collaboration
- Discussion forum
- Tech updates

Job/internship platform
- Job portal
- Internship portal
- Freelancing projects

2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] UpSkill Campus – Industrial Internship Project Guidelines and Problem Statement, *Forecasting of Smart City Traffic Patterns*.

[2] ChandniRana, *Smart City Traffic Management*, GitHub repository containing the traffic dataset and project resources.

[3] Scikit-learn Documentation – Machine Learning models and evaluation metrics used in the project.

2.6 Glossary

Terms	Acronym
Machine Learning	**ML**
Exploratory Data Analysis	**EDA**
Mean Absolute Error	**MAE**
Root Mean Squared Error	**RMSE**
Gradient Boosting	**GB**

3 Problem Statement

The project focuses on forecasting smart-city traffic patterns across four city junctions. The government aims to understand traffic variations and prepare for peak traffic periods to improve traffic management and future infrastructure planning.

The system analyzes traffic patterns on working days, weekends, holidays, and different times of the day. Machine Learning is used to forecast traffic and provide useful insights for better traffic management.

4 Existing and Proposed solution

5 Existing Solution

Traditional traffic management mainly depends on monitoring existing traffic conditions and responding to congestion when it occurs. Such approaches may have limited ability to forecast future traffic patterns and may not sufficiently consider variations caused by peak hours, weekends, and holidays.

6 Proposed Solution

The proposed system uses Data Science and Machine Learning to analyze historical traffic data from four city junctions and forecast traffic patterns. The system performs data preprocessing, feature engineering, exploratory analysis, and compares multiple Machine Learning models. Gradient Boosting was selected as the best-performing model with an R^2 score of 0.9691.

An interactive Streamlit dashboard was developed to visualize traffic patterns, identify busy junctions and peak hours, and provide traffic forecasts.

7 Value Addition

The project adds value by:

- Considering working days, weekends, and holidays.
- Identifying peak traffic periods.
- Analyzing traffic separately for four junctions.
- Using recent traffic history through lag and rolling features.
- Providing an interactive dashboard for easier interpretation.
- Supporting data-driven traffic management and infrastructure planning.

7.1 Code submission (Github link)

https://github.com/Rakshith-Kn/upskillcampus

7.2 Report submission (Github link) : first make placeholder, copy the link.

8 Proposed Design/ Model

The proposed system follows a step-by-step Machine Learning workflow. Historical traffic data is first collected and preprocessed. Relevant time-based, holiday, peak-hour, and traffic-history features are then created. Exploratory Data Analysis is performed to identify traffic patterns. Multiple Machine Learning models are trained and compared, and the best-performing Gradient Boosting model is selected for traffic forecasting. The final model is integrated into a Streamlit dashboard for analysis and prediction.

8.1 High Level Diagram

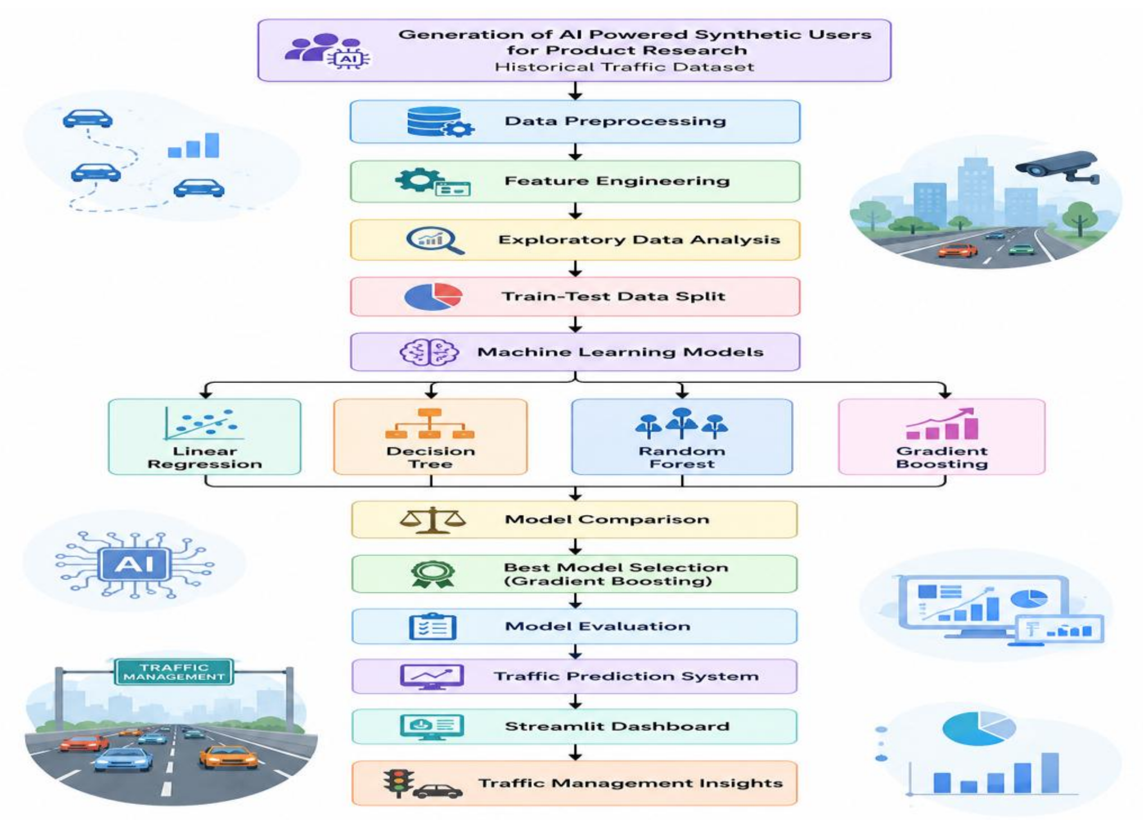
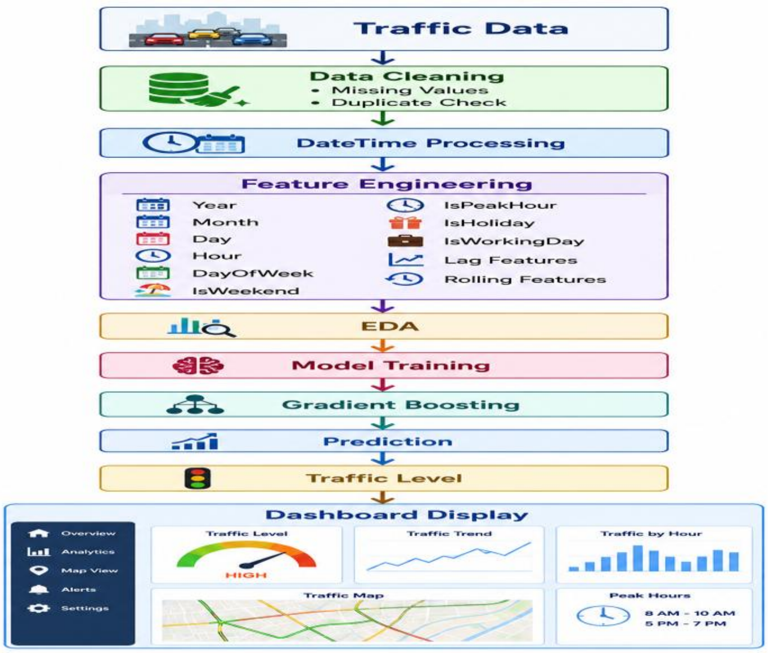


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

8.2 Low Level Diagram



8.3 Interfaces

The project provides an interactive Streamlit web interface for users.

The dashboard consists of three main sections:

1. Traffic Overview

Displays:

- Total traffic records
- Busiest junction
- Peak traffic hour
- Average traffic
- Traffic by day type

2. Traffic Pattern Analysis

Allows the user to select a junction and view:

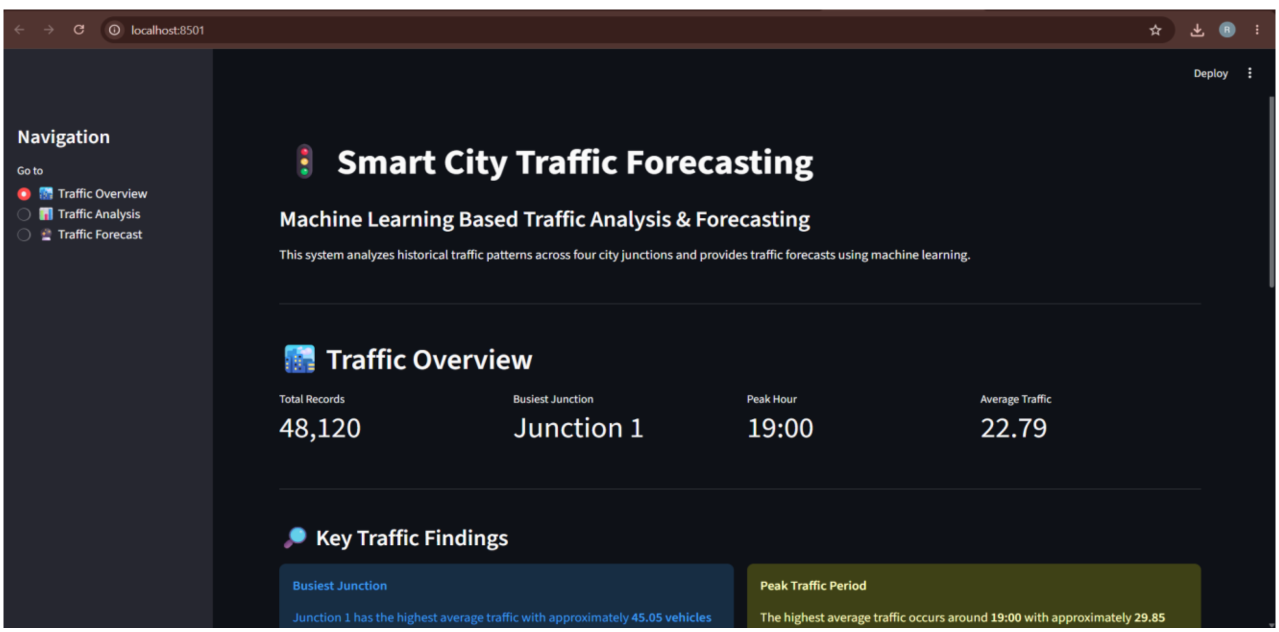
- Hourly traffic patterns
- Working-day, weekend and holiday traffic
- Holiday traffic patterns
- Peak-hour traffic

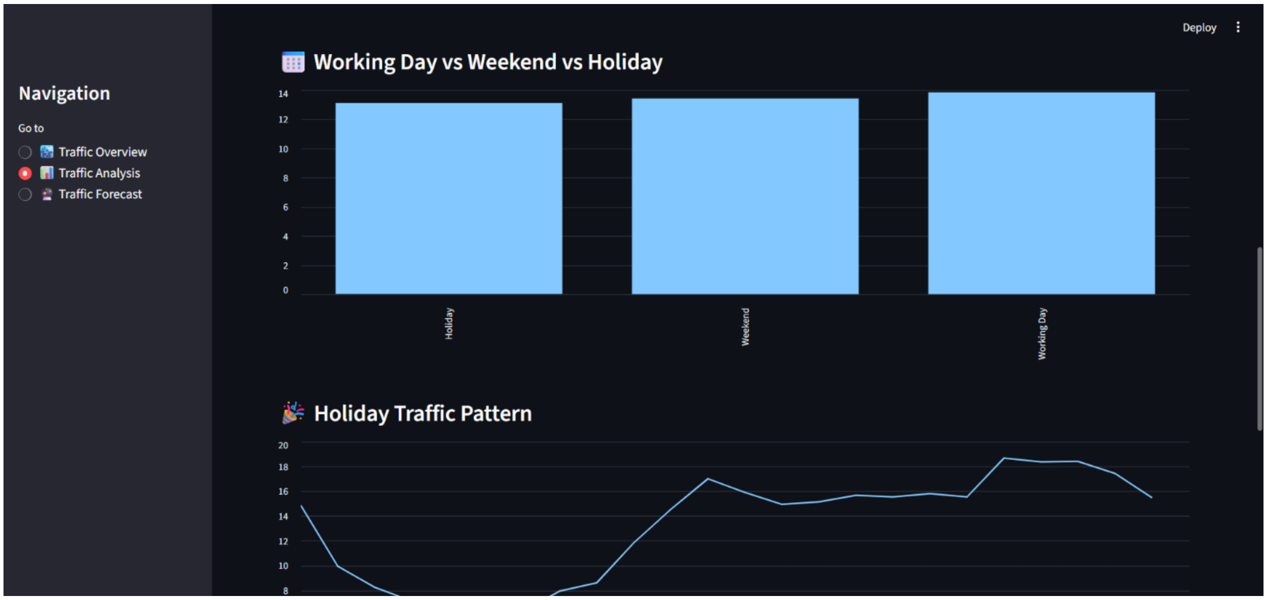
3.Traffic Forecast

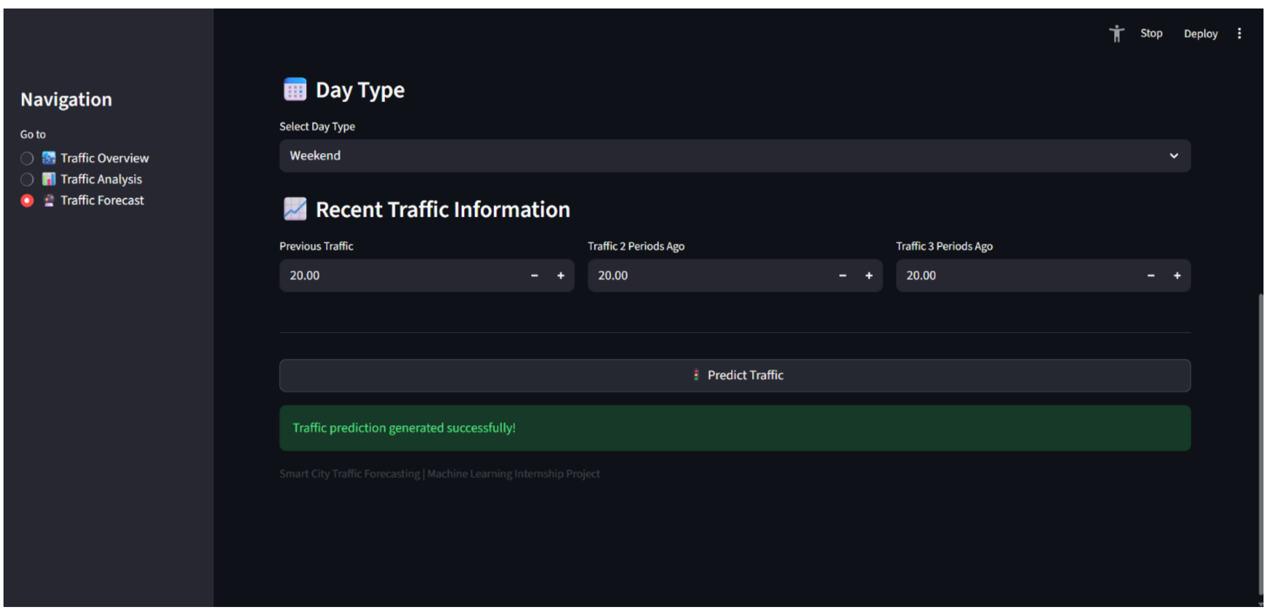
Allows the user to enter traffic-related parameters and generates:

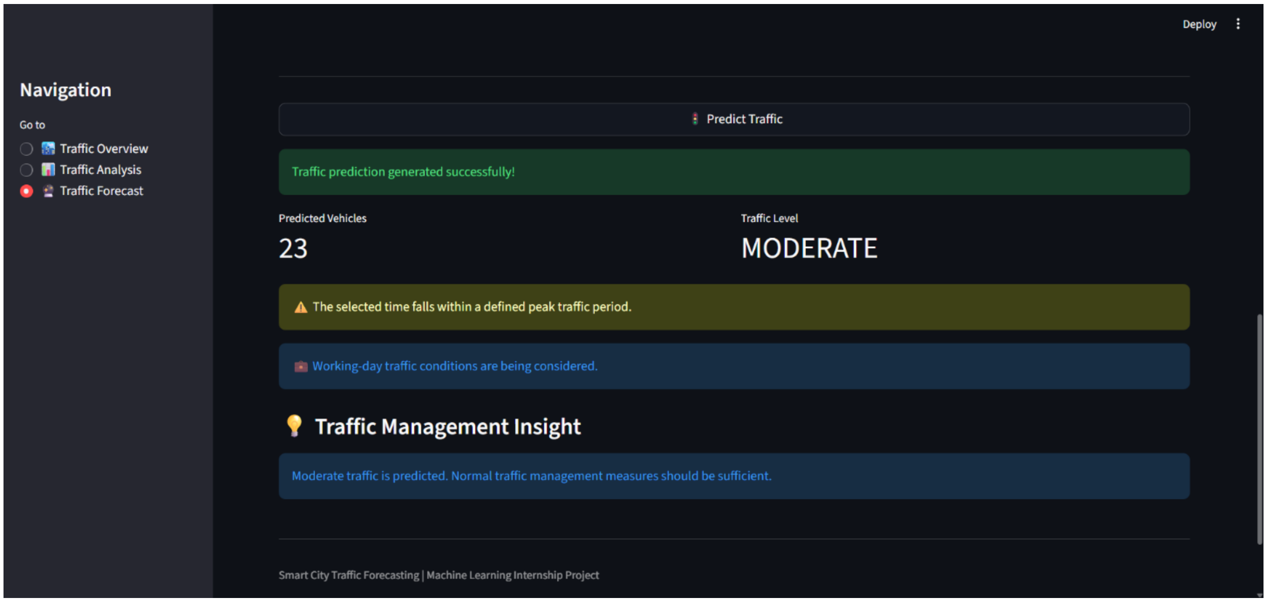
- Predicted number of vehicles
- Traffic level
- Peak-hour warning
- Traffic management recommendation

This interface makes the Machine Learning results easier to understand and use for traffic analysis.




Deploy
Working Day vs Weekend vs Holiday
Navigation
Go to
Traffic Overview
Traffic Analysis
Traffic Forecast
Holiday
Weekend
Working Day
Holiday Traffic Pattern


Stop
Deploy
Navigation
Go to
Traffic Overview
Traffic Analysis
Traffic Forecast
Day Type
Select Day Type
Weekend
Recent Traffic Information
Previous Traffic
20.00
Traffic 2 Periods Ago
20.00
Traffic 3 Periods Ago
20.00
Predict Traffic
Traffic prediction generated successfully!
Smart City Traffic Forecasting | Machine Learning Internship Project



9 Performance Test

The performance of the traffic forecasting system was evaluated using a time-based test dataset. The trained Machine Learning models were tested using historical traffic observations that were not used during training. The evaluation focused on prediction error and the ability of the model to represent traffic patterns during peak hours, holidays, and across different junctions.

9.1 Test Plan/ Test Cases

The system was tested at different stages to ensure that the traffic forecasting process worked correctly. The dataset was first checked for missing values and duplicate records, followed by verification of DateTime conversion and feature creation. The four Machine Learning models were then compared based on their prediction performance. The final model was tested on the unseen test data, with additional tests performed for peak-hour traffic, holiday traffic, and individual traffic junctions.

9.2 Test Procedure

A time-based train-test split was used, where earlier traffic records were used for training and later records were used for testing. Linear Regression, Decision Tree, Random Forest, and Gradient Boosting models were trained and evaluated using MAE, RMSE, and R^2 score. The selected model was further evaluated on peak-hour and holiday traffic, as well as separately for all four junctions.

9.3 Performance Outcome

The Gradient Boosting model achieved the best overall performance with an MAE of 2.9523, RMSE of 4.7814, and R^2 score of 0.9691. For peak-hour traffic, the model achieved an R^2 of 0.9695, while holiday traffic achieved an R^2 of 0.9735. The results show that the model can effectively learn historical traffic patterns and provide accurate traffic predictions.

10 My learnings

During this internship, I gained practical knowledge of Data Science and Machine Learning by working on a real-world traffic forecasting problem. I learned how to clean and preprocess data, perform feature engineering, analyze traffic patterns, train and compare Machine Learning models, and evaluate their performance using different metrics. I also gained experience in developing an interactive Streamlit dashboard and presenting Machine Learning results in a practical way. This experience improved my technical skills, problem-solving approach, and understanding of the complete Machine Learning workflow, which will be useful for my future career in software development and data-driven technologies.

11 Future work scope

In the future, the project can be enhanced by integrating real-time traffic data from sensors or IoT devices and using more advanced forecasting techniques. The system can also be improved to automatically generate future traffic predictions, provide real-time traffic alerts, and support better traffic management and infrastructure planning. Further improvements can include deploying the system on the cloud and expanding it to more city junctions.